

Übungsblatt 07: Visualization → SLAM

[Link to repo](#)

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Ros2 version. local 5.1.3

Prerequisites

I am using the pre installations of exercise 6. So I can use my windows PC and the already installed Docker Desktop.

Setup procedure

1. Start Docker Desktop, which is installed already
2. Run VS Code in my workspace folder of exercise 6 `lecture6-ros2demo`
3. As then VS Code prompts, *run in container*
4. Close the terminal, by pressing a button, as prompted.
5. Then in the Terminal check using `ros2 pkg list`
6. Some packages are missing, installing

```
root@49ed0b86c179:/workspace/turtlebot3_ws# sudo apt update && sudo apt install -y  
ros-humble-turtlebot3-cartographer ros-humble-nav2-map-server
```

5. Check packages present:

```
root@49ed0b86c179:/workspace/turtlebot3_ws# ros2 pkg list | grep -E  
"cartographer|map_server"  
cartographer_ros  
cartographer_ros_msgs  
nav2_map_server  
turtlebot3_cartographer  
root@49ed0b86c179:/workspace/turtlebot3_ws#
```

5. Check "Burger"

```
root@49ed0b86c179:/workspace/turtlebot3_ws# env | grep TURT  
TURTLEBOT3_MODEL=burger  
root@49ed0b86c179:/workspace/turtlebot3_ws#
```

5. Run commands according *Übungsblatt 7* in separate terminals:

- `ros2 launch turtlebot3_gazebo turtlebot3_world.launch.py`

- `ros2 launch turtlebot3_cartographer cartographer.launch.py - use_sim_time:=True`
- `ros2 run turtlebot3_teleop teleop_keyboard`
- `ros2 run nav2_map_server map_saver_cli -f ~/map`

Facing Problems

1. There was no `/workspace/turtlebot_ws`
2. The VNC server on `localhost:6080` did show the directory listing with `vnc-automation.html` which then started the linux desktop. (The Password was 1234). Expected where the gazebo GUI.

Solution

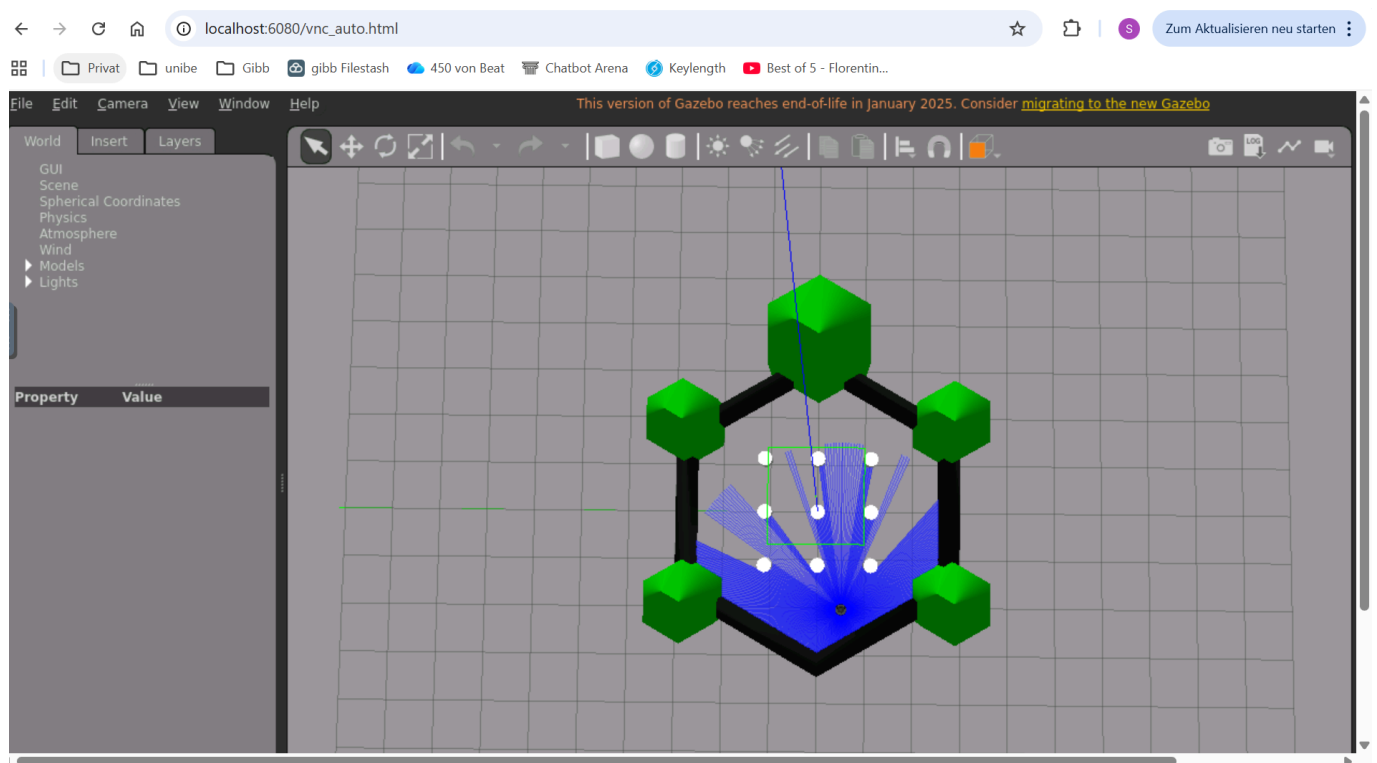
Important Stop containers in Docker Desktop!!

In VS Code , press F1 and run **Dev Containers: Rebuild Container**

Then close the terminal, reopen a new one and enter `pwd` which shows `/workspace/turtlebot3_ws`

And `echo $ROS_DISTRO` shows `humble`

Enter `tb3_world` then `localhost:6080` shows :



Hint CTRL+Shift+R does a hard refresh of the Browser, in case of errors like

Ein Fehler ist aufgetreten:

Cannot read properties of null (reading 'addEventListener') TypeError: Cannot read properties of null (reading '>'addEventListener') at Object.addClipboardHandlers (http://localhost:6080/app/ui.>js:329:13) at Object.start (http://localhost:6080/app/ui.js:101:12) at http://localhost:6080/app/ui.js:47:27

Check

```

root@e5e84fc1aa6b:/workspace/turtlebot3_ws# ros2 node list
/gazebo
/robot_state_publisher
/turtlebot3_diff_drive
/turtlebot3_imu
/turtlebot3_joint_state
/turtlebot3_laserscan
root@e5e84fc1aa6b:/workspace/turtlebot3_ws#

```

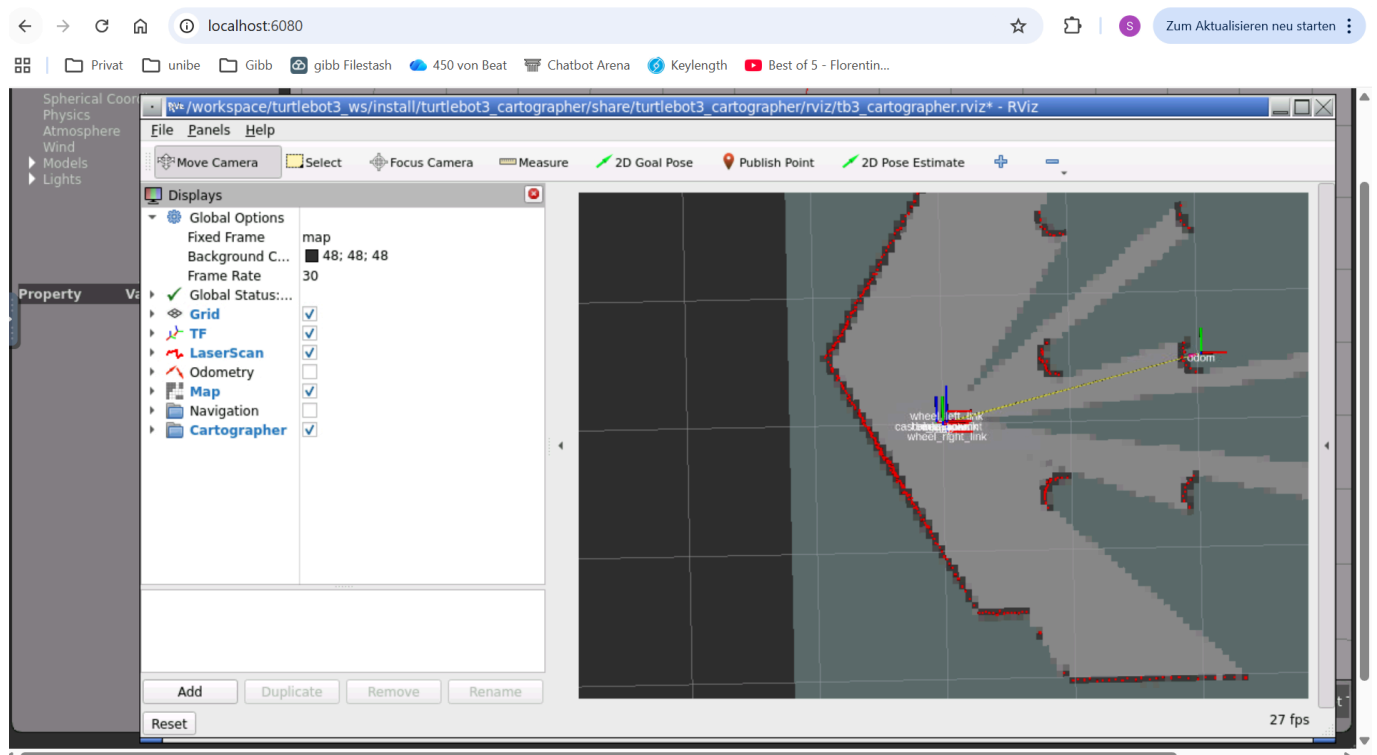
Instead of

- `ros2 launch turtlebot3_gazebo turtlebot3_world.launch.py` just run
- `tbs_world`

Now continue with

- `ros2 launch turtlebot3_cartographer cartographer.launch.py use_sim_time:=True`

This starts **Rviz**



Now start SLAM

1. `ros2 launch turtlebot3_cartographer cartographer.launch.py use_sim_time:=True`
2. `ros2 run turtlebot3_teleop teleop_keyboard`

Now we see the bot turning and moving according to the teleop command.

See the video [video teleop.mp4](#)

And moving around on the map: [moving around](#)

3. Save the map using again a new Terminal `ros2 run nav2_map_server map_saver_cli -f ~/map`

now we have

- `~map.yaml`
- `~map.pgm`

See the directory listing below:

```
root@e5e84fc1aa6b:~# ll -h | grep map
-rw-r--r-- 1 root root    0 May  3 08:06 .Xmodmap
-rw-r--r-- 1 root root 15K May  3 18:57 map.pgm
-rw-r--r-- 1 root root 121 May  3 18:57 map.yaml
drwxr-xr-x 2 root root 4.0K May  3 18:15 maps/
root@e5e84fc1aa6b:~#
```

What is going on

- The robot publishes laser scans (/scan)
- SLAM estimates position + environment simultaneously
- RViz visualizes the map (/map)
- Revisiting areas improves accuracy